

3F7: Information Theory and Coding

Handout 2: Law of Large Numbers, Typicality, Data Compression

Ramji Venkataramanan

Signal Processing and Communications Lab, CUED
rv285@cam.ac.uk

Michaelmas Term 2025

1 / 26

Estimating Tail Probabilities

We often want to bound the probability of rare/unlikely events. These correspond to the probability mass in the 'tails' of the pmf/density function.

Example: The number of cars passing through a street is Poisson distributed with mean $\lambda = 5/\text{minute}$. What is the probability that there are greater than 20 cars passing through in any given minute?

$$P(X > 20) = \sum_{r=21}^{\infty} P(X = r), \quad P(X = r) = e^{-\lambda} \lambda^r / r! \text{ for } r \geq 0$$

What if we do not know that the distribution is Poisson? Suppose we are only given that the average is 5/minute. Can we get a bound for $P(X > 20)$?

Markov and Chebyshev inequalities are ways to bound tail probabilities with limited information about the random variable.

- ▶ Markov is for non-negative rvs and requires only the mean
- ▶ Chebyshev is for general rvs, and requires mean and variance

2 / 26

Markov's Inequality

For a non-negative rv X and any $a > 0$,

$$P(X \geq a) \leq \frac{\mathbb{E}[X]}{a}.$$

► Note that it is useful only for $a > \mathbb{E}[X]$.

Example 1: Bound the probability that the number of cars passing through in a minute is greater than 20, given only that the average is 5/min.

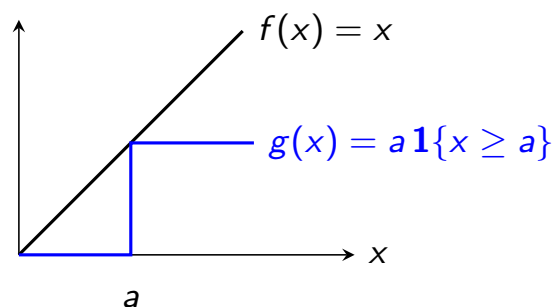
$$P(X > 20) = P(X \geq 21) \leq \frac{\mathbb{E}[X]}{21} = \frac{5}{21} = 0.2381$$

Example 2: The expected number of simultaneous calls at a cellular base station is 100. The station is designed to handle upto 150 simultaneous calls. Bound the probability that the station receives more than 150 simultaneous calls. (Ans: prob ≤ 0.662)

3 / 26

Proof of Markov's Inequality

Define the *indicator* function $\mathbf{1}\{x \geq a\} = \begin{cases} 1 & \text{if } x \geq a \\ 0 & \text{if } x < a \end{cases}$



Notice that for $x \geq 0$, we have $x \geq a \mathbf{1}\{x \geq a\}$. Therefore:

$$\sum_{x \geq 0} x P(X = x) \geq \sum_{x \geq 0} a \mathbf{1}\{x \geq a\} P(X = x) \quad (1)$$

The left side above equals $\mathbb{E}[X]$. The right side equals

$$a \sum_{x \geq 0} \mathbf{1}\{x \geq a\} P(X = x) = a \sum_{x \geq a} P(X = x) = a P(X \geq a)$$

Using this in (1), we obtain the result: $\mathbb{E}[X] \geq a P(X \geq a)$

4 / 26

Chebyshev's inequality

Often, we bound the tail probabilities of *deviations around the mean* of an rv, i.e., probabilities of the form

$$P(|X - \mathbb{E}X| \geq a), \text{ for some } a > 0.$$

Example: A circuit is designed to handle a nominal current of 20 mA plus or minus a deviation of less than 5 mA. If the applied current has mean 20 mA and variance 4 mA², bound the probability that the applied current violates the design parameters.

Denoting the applied current by X , what we want is

$$P(|X - \mathbb{E}X| \geq 5), \text{ with } \mathbb{E}X = 20 \text{ and } \text{Var}(X) = 4$$

Chebyshev's inequality is a way to bound the probability of deviations around the mean, with knowledge of just the *mean and variance*.

5 / 26

Chebyshev's inequality

For any rv X and $a > 0$,

$$P(|X - \mathbb{E}X| \geq a) \leq \frac{\text{Var}(X)}{a^2}$$

Applying Chebyshev to the example on prev. page, we get

$$P(|X - \mathbb{E}X| \geq 5) \leq \frac{\text{Var}(X)}{5^2} = 0.16.$$

Proof: Note that $P(|X - \mathbb{E}X| \geq a) = P(|X - \mathbb{E}X|^2 \geq a^2)$. Let $Y := |X - \mathbb{E}X|^2$.

Since Y is a non-negative rv, we can apply Markov's inequality to get

$$P(Y \geq a^2) \leq \frac{\mathbb{E}Y}{a^2} \Rightarrow P(|X - \mathbb{E}X| \geq a) \leq \frac{\text{Var}(X)}{a^2},$$

where we have used $\mathbb{E}Y = \mathbb{E}[(X - \mathbb{E}X)^2] = \text{Var}(X)$. □

6 / 26

- ▶ Markov and Chebyshev are nice to get bounds on the tail probability knowing just the mean, variance.
- ▶ However, these bounds are often loose, i.e, the upper bound may be much larger than the true probability. But that's the price you pay for knowing only the mean and the variance.
- ▶ There's another technique called the Chernoff bound which generally gives better bounds than Chebyshev/Markov (but it is based on more knowledge about the distribution than just the mean and variance). We won't discuss the Chernoff bound in this course.

We now use Chebyshev to establish the **law of large numbers**.

7 / 26

Weak Law of Large Numbers (WLLN)

Roughly: "*Empirical average converges to the mean*"

Let $X_1, X_2, \dots$ be a sequence of i.i.d. random variables with finite mean μ . Let $S_n = \frac{1}{n} \sum_{i=1}^n X_i$.

- ▶ *Informal statement of WLLN:* $S_n \rightarrow \mu$ as $n \rightarrow \infty$.
- ▶ *Formal statement of WLLN:* For any $\epsilon > 0$, $\lim_{n \rightarrow \infty} P(|S_n - \mu| \geq \epsilon) = 0$.

Proof: We will prove the result assuming that the variance of each X_i is finite, say σ^2 . By Chebyshev's inequality, we have

$$P(|S_n - \mu| \geq \epsilon) \leq \frac{\text{Var}(S_n)}{\epsilon^2} = \frac{\sigma^2}{n\epsilon^2}. \quad (2)$$

The last equality above is obtained using the independence of the X_i 's:

$$\text{Var}(S_n) = \frac{1}{n^2} \text{Var}\left(\sum_i X_i\right) = \frac{1}{n^2} \sum_{i=1}^n \text{Var}(X_i) = \frac{1}{n^2} n\sigma^2 = \frac{\sigma^2}{n}.$$

For any fixed $\epsilon > 0$, note that $\frac{\sigma^2}{n\epsilon^2} \rightarrow 0$ as $n \rightarrow \infty$. Hence (2) implies $\lim_{n \rightarrow \infty} P(|S_n - \mu| \geq \epsilon) = 0$. □

8 / 26

A few remarks on the WLLN:

- ▶ Note that ϵ can be arbitrarily small (e.g., $\epsilon = 10^{-8}$) but is held *fixed* as n tends to ∞ .
- ▶ Example: Let $X_1, \dots, X_n$ be Bernoulli rvs with $P(X_i = 1) = 1 - P(X_i = 0) = \frac{1}{4}$. Then S_n is the fraction of 1's observed in n trials. WLLN says that the probability of this fraction deviating from $\frac{1}{4}$ by more than ϵ goes to zero as $n \rightarrow \infty$.

Exercise: According the Chebyshev bound in (2) above, what is the minimum number of trials needed for the deviation probability to be less than 0.01 for $\epsilon = 10^{-3}$?

Ans: $P(|S_n - \mu| \geq \epsilon) \leq \frac{\sigma^2}{n\epsilon^2} < 0.01$. This implies that $n > \frac{\sigma^2}{(0.01)\epsilon^2}$. For the Bernoulli rv, the variance $\sigma^2 = \frac{1}{4} \frac{3}{4} = \frac{3}{16}$.

- ▶ The WLLN is an example of *convergence in probability* of a sequence of random variables: We say that the sequence of rvs $\{S_n\}_{n=1,2,\dots}$ converges in probability to the expectation μ .
- ▶ As we will see in this course, much of information theory is a (clever) application of WLLN!

9 / 26

Typicality – a simple example

Consider an i.i.d. Bernoulli($\frac{1}{4}$) source. It produces symbols $X_1, X_2, \dots$ according to

$$P(X_i = 1) = \frac{1}{4}, \quad P(X_i = 0) = \frac{3}{4} \quad \text{for } i = 1, 2, \dots$$

One of the following sequences is a “real” output of the source.

1. 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2. 0 1 0 0 0 0 0 0 1 1 0 0 0 1 0 0
3. 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

Which sequence is a real output?

1. 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2. 0 1 0 0 0 0 0 0 1 1 0 0 0 1 0 0
3. 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1

- ▶ Each sequence has 16 bits.
- ▶ The probability of a sequence with k ones and $16 - k$ zeros is

$$\left(\frac{1}{4}\right)^k \left(\frac{3}{4}\right)^{16-k}$$

- ▶ Probability of source emitting first sequence = $\left(\frac{3}{4}\right)^{16}$
- ▶ Probability of source emitting second sequence = $\left(\frac{1}{4}\right)^4 \left(\frac{3}{4}\right)^{12}$

The *first* sequence is 3^4 times more likely than the second!

11 / 26

Typical sequences

1. 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0
2. 0 1 0 0 0 0 0 0 1 1 0 0 0 1 0 0

- ▶ Though less likely than the first sequence, the second sequence seems more “typical” of the $\left(\frac{1}{4}, \frac{3}{4}\right)$ source.
- ▶ We will make this idea precise via the notion of a typical set.

If $X_1, \dots, X_n$ are chosen $\sim$ i.i.d. Bernoulli(p), then for large n :

- ▶ With high probability, the fraction of ones in the observed sequence will be close to p (due to WLLN).
- ▶ Equivalently: with high probability, the observed sequence will have probability close to $p^{np}(1-p)^{n(1-p)}$
- ▶ Note that any number a can be written as $2^{\log a}$. Hence

$$p^{np}(1-p)^{n(1-p)} = (2^{\log p})^{np} (2^{\log(1-p)})^{n(1-p)} = 2^{-nH_2(p)}$$

For large n , with high probability we will observe a *typical* sequence. Informally, a typical sequence is one whose probability is close to $2^{-nH_2(p)}$.

12 / 26

Asymptotic Equipartition Property (AEP)

The observation above gives an *operational* meaning to entropy

The AEP makes this precise for any i.i.d. discrete source, not just Bernoulli sequences.

Asymptotic Equipartition Property (AEP)

If $X_1, X_2, \dots$ are i.i.d. $\sim P_X$, then for any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \Pr \left(\left| \frac{-1}{n} \log P_X(X_1, X_2, \dots, X_n) - H(X) \right| < \epsilon \right) = 1.$$

Remarks:

- ▶ $\frac{-1}{n} \log P_X(X_1, X_2, \dots, X_n)$ is a *random variable*
(Note the capitals; A function of the rvs $(X_1, \dots, X_n)$ is a rv)
- ▶ AEP says this rv converges in probability to $H(X)$, a **constant**, as $n \rightarrow \infty$.

13 / 26

Proof of the AEP

Simple application of Weak Law of Large Numbers (WLLN). Let

$$Y_i = -\log P_X(X_i), \quad \text{for } i = 1, \dots, n.$$

- ▶ *Functions of independent rvs are also independent rvs*
 $\Rightarrow Y_1, \dots, Y_n$ are i.i.d.
- ▶ *WLLN for Y_i 's* says that for any $\epsilon > 0$

$$\lim_{n \rightarrow \infty} \Pr \left(\left| \frac{1}{n} \sum_i Y_i - \mathbb{E}[Y_1] \right| < \epsilon \right) = 1. \quad (3)$$

- ▶ Note that

$$\begin{aligned} \sum_i Y_i &= -\sum_i \log P_X(X_i) = -\log [P_X(X_1)P_X(X_2) \dots P_X(X_n)] \\ &\stackrel{\text{why?}}{=} -\log P_X(X_1, X_2, \dots, X_n) \end{aligned} \quad (4)$$

- ▶ Substitute (4) in (3), and note that $\mathbb{E}[Y_1] = H(X)$ to get the AEP.

□

14 / 26

The Typical Set

Recall the AEP:

If $X_1, X_2, \dots$ are i.i.d. $\sim P$, then for any $\epsilon > 0$

$$\Pr \left(\left| -\frac{1}{n} \log P(X_1, X_2, \dots, X_n) - H(X) \right| < \epsilon \right) \xrightarrow{n \rightarrow \infty} 1.$$

The **typical set** $A_{\epsilon, n}$ with respect to P is the set of sequences $(x_1, \dots, x_n) \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \leq P(x_1, \dots, x_n) \leq 2^{-n(H(X)-\epsilon)}$$

“Sequences whose probability is concentrated around $2^{-nH(X)}$ ”

- Note the dependence on n and ϵ
- A sequence belonging to the typical set is called an ϵ -typical sequence.

15 / 26

Properties of the Typical Set

Notation: We will use X^n to denote the vector $X_1, X_2, \dots, X_n$

Property 1

If $X^n = (X_1, \dots, X_n)$ is generated i.i.d. $\sim P$, then

$$\Pr(X^n \in A_{\epsilon, n}) \xrightarrow{n \rightarrow \infty} 1$$

That is, the sequence $X_1, \dots, X_n$ you observe is very likely to belong to the typical set.

Proof: From the definition of $A_{\epsilon, n}$, note that

$$\begin{aligned} X^n \in A_{\epsilon, n} &\Leftrightarrow 2^{-n(H(X)+\epsilon)} \leq P(X^n) \leq 2^{-n(H(X)-\epsilon)} \\ &\Leftrightarrow H(X) - \epsilon \leq -\frac{1}{n} \log P(X^n) \leq H(X) + \epsilon \end{aligned} \quad (5)$$

AEP says that

$$\Pr(X^n \in A_{\epsilon, n}) = \Pr \left(H(X) - \epsilon \leq -\frac{1}{n} \log P(X^n) \leq H(X) + \epsilon \right) \xrightarrow{n \rightarrow \infty} 1.$$

□

16 / 26

Property 2

Let $|A_{\epsilon,n}|$ denote the number of elements in the typical set $A_{\epsilon,n}$. Then

$$|A_{\epsilon,n}| \leq 2^{n(H(X)+\epsilon)}$$

Proof:

$$\begin{aligned} 1 &= \sum_{x^n \in \mathcal{X}^n} P(x^n) \\ &\geq \sum_{x^n \in A_{\epsilon,n}} P(x^n) \\ &\stackrel{(a)}{\geq} \sum_{x^n \in A_{\epsilon,n}} 2^{-n(H(X)+\epsilon)} \\ &= 2^{-n(H(X)+\epsilon)} |A_{\epsilon,n}| \end{aligned}$$

Hence $|A_{\epsilon,n}| \leq 2^{n(H(X)+\epsilon)}$. (Inequality (a) follows from the definition of the typical set.) $\square$

17 / 26

Property 3

For sufficiently large n , $|A_{\epsilon,n}| \geq (1 - \epsilon) 2^{n(H(X)-\epsilon)}$

Proof: From Property 1, $Pr(X^n \in A_{\epsilon,n}) \rightarrow 1$ as $n \rightarrow \infty$.

This means that for any $\epsilon > 0$, for sufficiently large n we have $Pr(X^n \in A_{\epsilon,n}) > 1 - \epsilon$. Thus, for sufficiently large n :

$$\begin{aligned} 1 - \epsilon &< Pr(X^n \in A_{\epsilon,n}) \\ &= \sum_{x^n \in A_{\epsilon,n}} P(x^n) \\ &\stackrel{(b)}{\leq} \sum_{x^n \in A_{\epsilon,n}} 2^{-n(H(X)-\epsilon)} \\ &= 2^{-n(H(X)-\epsilon)} |A_{\epsilon,n}| \end{aligned}$$

Hence $|A_{\epsilon,n}| \geq (1 - \epsilon) 2^{n(H(X)-\epsilon)}$. (Inequality (b) follows from the definition of the typical set.) $\square$

18 / 26

Properties of the Typical Set

Summary: For large n ,

- ▶ Suppose you generate $X_1, \dots, X_n$ i.i.d. $\sim P$. With high probability, the sequence you obtain will be typical, i.e., its probability is close to $2^{-nH(X)}$.
- ▶ If the pmf P assigns non-zero probabilities to the symbols denoted a, b, c etc., the typical set is essentially the set of sequences whose fraction of a 's is close to $P(a)$, fraction of b 's is close to $P(b)$, and so on.
- ▶ The number of typical sequences is close to $2^{nH(X)}$.

We will now see how all this is relevant to data compression.

19 / 26

Compression

GOAL: To compress a source producing symbols $X_1, X_2, \dots \in \mathcal{X}$ that are i.i.d. $\sim P$.

For any n , a compression code is defined as follows:

- ▶ To each source sequence $X^n := (X_1, \dots, X_n)$, the code assigns a *unique* binary sequence $c(X^n)$.
- ▶ $c(X^n)$ is called the *codeword* for the source sequence X^n .
- ▶ Let $\ell(X^n)$ be the length of the codeword assigned to X^n , i.e., the number of bits in $c(X^n)$.
- ▶ The expected code length is defined as

$$\mathbb{E}[\ell(X^n)] = \sum_{x^n} P(x^n) \ell(x^n).$$

The goal is to come up with a code (a mapping) that has small expected code length — the smaller the better!

A natural question is: what is the shortest expected code length possible for a given source distribution P ?

20 / 26

For concreteness, let us consider English text as an example:

$$\mathcal{X} = \{a \ b \ \dots \ z \ , \ . \ \text{space} \ ; \ @ \ \#\} \quad |\mathcal{X}| = 32$$

(English text is not really i.i.d., but for now assume it is)

- ▶ Assume that we know the source entropy $H(X)$.
- ▶ $H(X)$ can be estimated by measuring the frequency of each symbol. E.g. by measuring the frequencies of a, b etc. separately, the entropy estimate for English text is ≈ 4 bits.



21 / 26

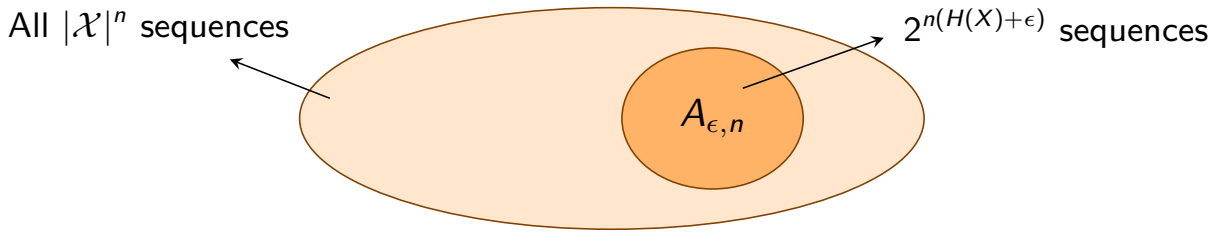
A Naïve Compression Code

- ▶ List all the $|\mathcal{X}|^n$ possible length n sequences.
- ▶ Index these as $\{0, 1, \dots, |\mathcal{X}|^n - 1\}$ using $\lceil \log |\mathcal{X}|^n \rceil$ bits $\Rightarrow$ Length of codeword for each sequence = $n \log |\mathcal{X}|$ (5n for English)
- ▶ Expected code length: $\mathbb{E}[\ell(X^n)] = n \log |\mathcal{X}|$
- ▶ Expected number of bits/symbol: $\mathbb{E}[\ell(X^n)]/n = \log |\mathcal{X}|$ (5 bits/symbol for English)

How can we do better?

22 / 26

Compression via the Typical Set



Compression scheme:

- There are at most $2^{n(H(X)+\epsilon)}$ ϵ -typical sequences. ($2^{n(4+\epsilon)}$ for our example, since $H(X) = 4$)
- Index each sequence in $A_{\epsilon, n}$ using $\lceil \log 2^{n(H(X)+\epsilon)} \rceil$ bits. Prefix each of these by a flag bit 0.

$$\text{Bits/typical seq.} = \lceil n(H(X) + \epsilon) \rceil + 1 \leq n(H(X) + \epsilon) + 2$$

- Index each sequence *not* in $A_{\epsilon, n}$ using $\lceil \log |\mathcal{X}|^n \rceil$ bits. Prefix each of these by a flag bit 1.

$$\text{Bits/non-typical seq.} = \lceil n \log |\mathcal{X}| \rceil + 1 \leq n \log |\mathcal{X}| + 2$$

This code assigns a *unique* codeword to each sequence in $|\mathcal{X}|^n$

23 / 26

Expected code length

Let $\ell(X^n)$ be length of the codeword assigned to sequence X^n .

$$\begin{aligned} \mathbb{E}[\ell(X^n)] &= \sum_{x^n} P(x^n) \ell(x^n) \\ &= \sum_{x^n \in A_{\epsilon, n}} P(x^n) \ell(x^n) + \sum_{x^n \notin A_{\epsilon, n}} P(x^n) \ell(x^n) \\ &\leq \sum_{x^n \in A_{\epsilon, n}} P(x^n) (n(H(X) + \epsilon) + 2) + \sum_{x^n \notin A_{\epsilon, n}} P(x^n) (n \log |\mathcal{X}| + 2) \\ &\leq 1 \cdot n(H(X) + \epsilon) + \epsilon \cdot n \log |\mathcal{X}| + 2 \left(\sum_{x^n \in A_{\epsilon, n}} P(x^n) + \sum_{x^n \notin A_{\epsilon, n}} P(x^n) \right) \\ &= n(H(X) + \epsilon) + \epsilon n \log |\mathcal{X}| + 2 \\ &= n(H(X) + \epsilon') \end{aligned}$$

where $\epsilon' = \epsilon + \epsilon \log |\mathcal{X}| + \frac{2}{n}$.

ϵ' can be made arbitrarily small by picking ϵ small enough and then n sufficiently large.

24 / 26

Fundamental Limit of Compression

We have just shown that we can represent sequences X^n using $nH(X)$ bits on average. More precisely, we have shown the following result.

Let X^n be i.i.d. $\sim P$. Fix any $\epsilon > 0$. For n sufficiently large, there exists a code that maps sequences x^n of length n into binary strings such that the mapping is *one-to-one* and

$$\mathbb{E} \left[\frac{1}{n} \ell(X^n) \right] \leq H(X) + \epsilon.$$

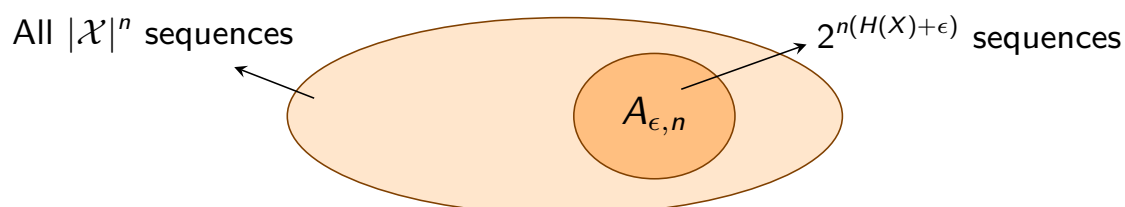
In fact, more is true — you *cannot* do better than $H(X)$, i.e., the expected length of *any* uniquely decodable code satisfies

$$\mathbb{E} \left[\frac{1}{n} \ell(X^n) \right] \geq H(X)$$

(We will prove a weaker version of this statement in the next handout.)

Hence entropy is the fundamental limit of lossless compression

25 / 26



Summary:

- ▶ For a non-uniform source, the typical set is a *small* subset of the set of all sequences, but contains almost all the prob.!
- ▶ This is the reason that even with an i.i.d. model, English text can be compressed to ≈ 4 bits/sample.
- ▶ Can compress much more if we consider *dependencies* in the text. E.g. q always followed by u .
- ▶ What kind of source cannot be compressed at all ?

Is scheme described above *practical*?

- No. To find the codeword for any x^n , we need to go through a table of 2^{nH} entries – *computationally complex*!
- We will next look at some practical schemes with much lower complexity, and examine how close they get to entropy.

You can now do up to question 9 on Examples Paper 1.

26 / 26